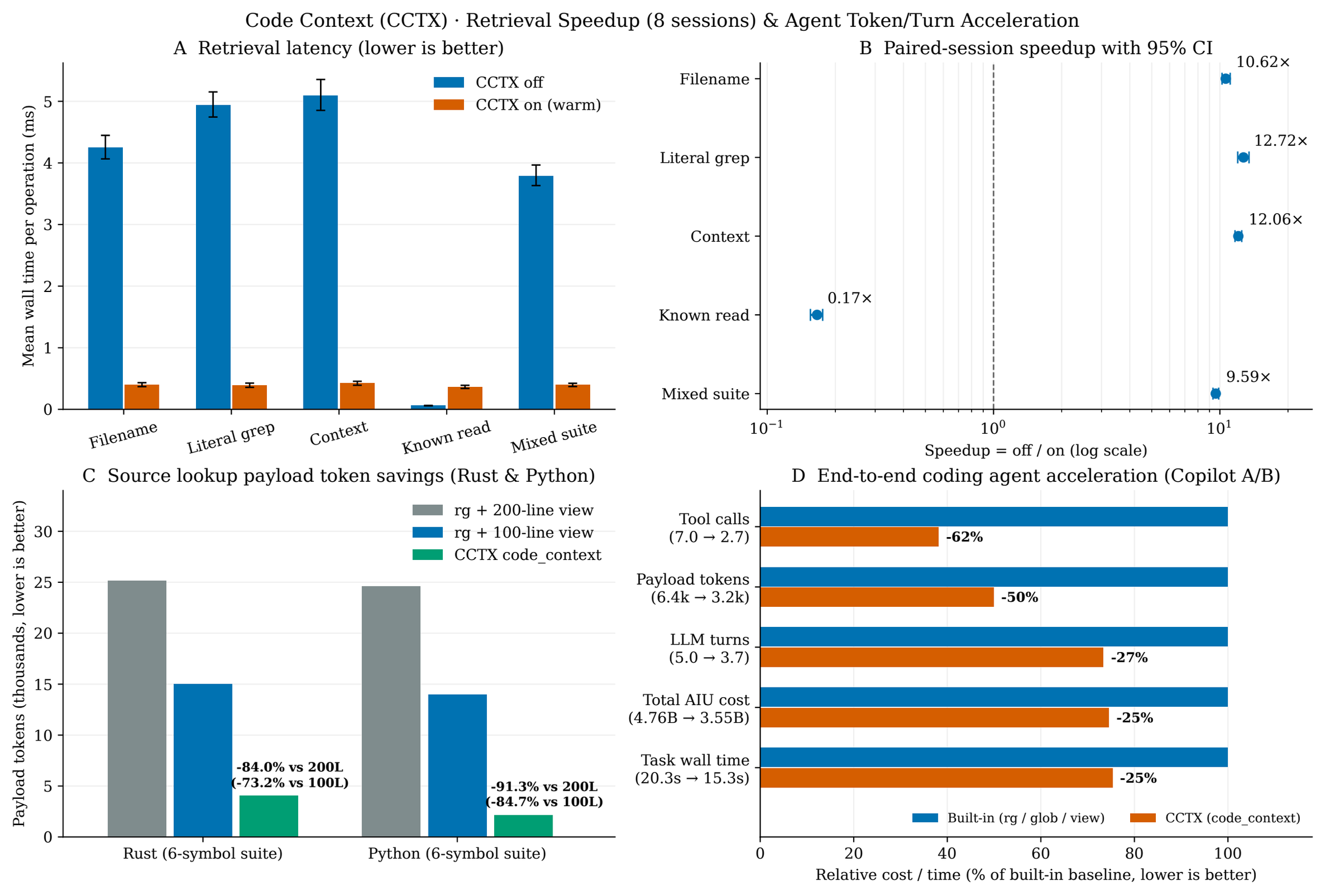
Code Context (CCTX) · Retrieval Speedup (8 sessions) & Agent Token/Turn Acceleration
A Retrieval latency (lower is better)
Mean wall time per operation (ms)
CCTX off
CCTX on (warm)
Filename
Literal grep
Context
Known read
Mixed suite
B Paired-session speedup with 95% CI
10.62×
12.72×
12.06×
0.17×
9.59×
Speedup = off / on (log scale)
C Source lookup payload token savings (Rust & Python)
Payload tokens (thousands, lower is better)
rg + 200-line view
rg + 100-line view
CCTX code_context
-84.0% vs 200L
(-73.2% vs 100L)
-91.3% vs 200L
(-84.7% vs 100L)
Rust (6-symbol suite)
Python (6-symbol suite)
D End-to-end coding agent acceleration (Copilot A/B)
Tool calls
(7.0 → 2.7)
-62%
Payload tokens
(6.4k → 3.2k)
-50%
LLM turns
(5.0 → 3.7)
-27%
Total AIU cost
(4.76B → 3.55B)
-25%
Task wall time
(20.3s → 15.3s)
-25%
Built-in (rg / glob / view)
CCTX (code_context)
Relative cost / time (% of built-in baseline, lower is better)